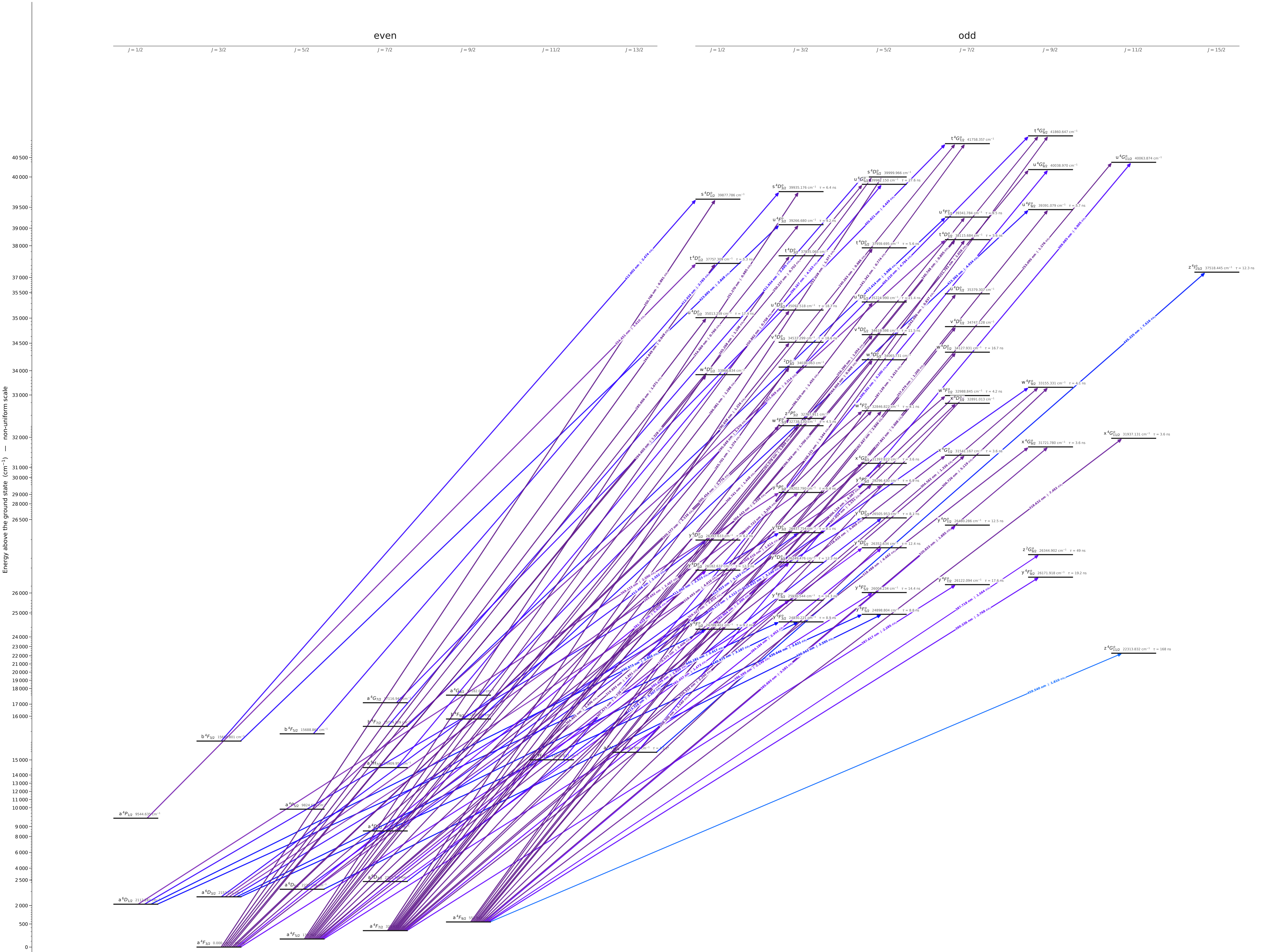


V<sup>+</sup> ionisation limit 54411.67 cm<sup>-1</sup> = 6.74619 eV (183.784 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|\langle f | e r | f' \rangle|$  in  $e a_0$   
 Matrix elements are measured or NIST values; \* = high-accuracy theory,  $\approx$  = model calculation.  
 Arrow width grows with the Einstein coefficient  $A$ . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists)



| Transition                | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $ U e J J $ (ea0) | A (s <sup>-1</sup> ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use           |
|---------------------------|--------------------|--------------------|-------------|-------------------|----------------------|-------------------------|------------|---------------|
| $a^4H_{132} - z^4F_{15Q}$ | 445.3255           | 445.2005           | 673.1985    | 7.516             | 8.100e+07            | 12.94                   | 100        | laser-cooling |

$\lambda, \nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
 $\Gamma/2\pi$ : natural linewidth of the upper level, 1/(2 $\pi\tau$ ).

| Level      | A (MHz) | B (MHz) | Shift of each F level from the centre of gravity (MHz)                                                          |
|------------|---------|---------|-----------------------------------------------------------------------------------------------------------------|
| $y^0_{F5}$ | 9930    | -       | F=1: -1112.98 F=2: -915.12 F=3: -618.32 F=4: -222.60 F=5: +272.06 F=6: +865.65                                  |
| $y^0_{D5}$ | 1530    | -       | F=1: -168.63 F=2: -138.65 F=3: -93.69 F=4: -33.73 F=5: +41.22 F=6: +131.16                                      |
| $y^0_{F6}$ | 54060   | -       | F=2: -3642.48 F=3: -2023.60 F=4: +134.91 F=5: +2833.04                                                          |
| $y^0_{F6}$ | 39060   | -       | F=3: -4381.48 F=4: -3605.09 F=5: -2346.81 F=6: +876.79 F=7: +1071.76 F=8: +3410.14                              |
| $x^0_{F6}$ | 60310   | -       | F=1: -6779.06 F=2: -5733.89 F=3: -2765.14 F=4: -1355.81 F=5: +1657.10 F=6: +5272.60                             |
| $x^0_{F7}$ | 14500   | -       | F=2: -4580.08 F=3: -3707.69 F=4: -2579.29 F=5: -1677.09 F=6: +367.09 F=7: +1526.69 F=8: +3562.29                |
| $x^0_{F7}$ | 15900   | -       | F=1: -3058.63 F=2: -2740.65 F=3: -264.18 F=4: -1628.62 F=5: -83.14 F=6: -119.17 F=7: +1231.40 F=8: +2502.52     |
| $x^0_{F8}$ | 81100   | -       | F=2: -1841.48 F=3: -1596.64 F=4: -1274.87 F=5: -870.75 F=6: -384.48 F=7: +182.12 F=8: +829.68 F=9: +1558.17     |
| $w^0_{F8}$ | 25800   | -       | F=1: -2900.50 F=2: -2384.85 F=3: -1611.34 F=4: -580.10 F=5: +709.01 F=6: +2255.94                               |
| $w^0_{F8}$ | 16900   | -       | F=2: -1699.82 F=3: -1591.89 F=4: -1376.04 F=5: -1052.27 F=6: -620.57 F=7: -60.94 F=8: +566.61 F=9: +1322.08     |
| $z^0_{F8}$ | 18130   | -       | F=4: -5440.47 F=5: -4526.11 F=6: -3428.87 F=7: -2148.76 F=8: -685.77 F=9: +960.08 F=10: +2788.81 F=11: +4800.42 |
| $z^0_{F9}$ | 21310   | -       | F=2: -1436.76 F=3: -798.20 F=4: +53.21 F=5: +1117.48                                                            |
| $t^0_{D9}$ | 14450   | -       | F=0: -2266.43 F=1: -2122.53 F=2: -1834.72 F=3: -1403.03 F=4: -827.42 F=5: -107.92 F=6: +755.48 F=7: +1467.85    |
| $u^0_{F9}$ | 28530   | -       | F=2: -5482.46 F=3: -4912.85 F=4: -4058.44 F=5: -2919.23 F=6: -1495.22 F=7: +213.60 F=8: +2207.22 F=9: +7742.78  |

Measured A, B constants; sources in the data file